

Arjun Mehta

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EDUCATION

Imperial College London | *MSc Computer Science (Distributed Systems)* 2019-2021

- Thesis on consensus algorithms achieved 98.7% Byzantine fault tolerance; published in IEEE Transactions on Parallel and Distributed Systems with 127 citations
- Graduated with Distinction (GPA 3.95/4.0); awarded the John Pratt Prize for outstanding research in systems engineering

ETH Zurich | *BSc Computer Science* 2015-2019

- Ranked top 2% of cohort; led Systems Theory lab as student mentor to 8 junior students over 2 years
- Completed internship on sharded database optimization reducing query latency by 62% (840ms → 320ms) for 50M+ records

EXPERIENCE

Stripe | *Senior Backend Engineer (Distributed Systems)* 2021-2024

- Architected and deployed a new transaction settlement service handling 500K+ TPS across 4 datacenters; reduced p99 latency from 145ms to 28ms (81% improvement) and achieved 99.999% uptime
- Led design and implementation of a distributed rate-limiting system protecting payment infrastructure; processed 2.3B requests/day with < 1ms overhead per request, preventing estimated \$12M+ in fraud losses
- Managed team of 5 engineers on consensus layer refactoring; reduced database replication lag from 3.2s to 180ms (94% improvement) across 6 sharded clusters, directly enabling new real-time product features

Cloudflare | *Staff Backend Engineer* 2018-2021

- Designed and shipped edge cache invalidation system using gossip protocol; reduced stale content serving by 99.2% with zero false positives across 180+ datacenters serving 250M requests/sec
- Optimized distributed tracing system reducing storage costs by \$2.8M annually while improving trace collection completeness from 87% to 99.6% across 50K+ monitored services
- Drove refactoring of HTTP/3 connection pooling logic; reduced memory footprint by 53% per connection (4.2MB → 1.9MB) enabling 35% more concurrent connections per edge server

Wise (formerly TransferWise) | *Backend Engineer* 2016-2018

- Built real-time foreign exchange distributed ledger system achieving 4ms 99th percentile latency across 8 regions; reconciliation completeness increased from 94% to 99.97% across \$140B+ in monthly transaction volume
- Implemented sharded event sourcing architecture reducing write contention by 71%; throughput increased from 45K ops/sec to 186K ops/sec supporting 2.1M daily active users
- Engineered failover orchestration system reducing MTTR from 8.5 minutes to 34 seconds; enabled fully automated recovery for 12 critical services with zero manual intervention

PROJECTS

ConsensusDB | *Rust, gRPC, Raft, Protocol Buffers* 2022-2024

- Open-source distributed database implementation achieving 2.1K GitHub stars; benchmarks show 94K writes/sec with sub-100ms replication latency across 5-node cluster
- Built comprehensive test suite with 847 unit tests covering Byzantine fault tolerance scenarios; achieved 99.4% code coverage and 2.3M downloads on crates.io

ShardMaster | *Go, etcd, Kubernetes, Protocol Buffers* 2021-2023

- Distributed sharding orchestration framework reducing data rebalancing time by 68% (from 45 minutes to 14 minutes) for clusters with 1.2TB+ data; prevents shard hotspots with 94% accuracy

- 1.4K GitHub stars; deployed in production across 8 companies; supports automatic replication factor adjustment maintaining 99.95% availability SLA

LatticeTrace | *Rust, Apache Kafka, ClickHouse, WebAssembly* 2020-2022

- Observability platform for distributed systems processing 180M spans/day with < 800µs processing latency; uses lazy evaluation to reduce storage requirements by 76% vs traditional APM
- 870 GitHub stars; successfully ingests traces from 340+ microservices; query response times improved by 91% (2.3s → 210ms avg) through columnar aggregation

ReplicaSync | *Scala, Apache Flink, Cassandra, gRPC* 2019-2021

- Real-time data replication engine supporting 5 consistency models with adaptive failover; 680 GitHub stars and used by 4 production teams handling 890K events/sec
- Achieved 99.98% replication accuracy with < 450ms max propagation delay across multi-region clusters; reduced operational incidents by 87% vs legacy replication approach

EXTRACURRICULAR

DistributedSystems.fyi | *Founder & Content Lead* 2021-Present

- Built educational platform with 240+ technical articles on consensus, sharding, and fault tolerance; accumulated 47K monthly readers and 8.2K newsletter subscribers
- Mentored 34 engineers through intensive system design program; 92% of mentees secured senior+ positions at tier-1 companies within 6 months

VLDB & ACM SIGMOD | *Technical Speaker & Conference Committee* 2020-Present

- Delivered talks on 'Consensus at Scale' (VLDB 2023) and 'Distributed Tracing Trade-offs' (SIGMOD 2022); combined audience reach of 2,400+ researchers; talks received average 4.7/5 feedback rating
- Reviewed 28 papers as committee member for distributed systems track; co-organized workshop on 'Byzantine Fault Tolerance in Production' attended by 180+ practitioners

TECHNICAL SKILLS

Languages – Rust, Go, Scala, Python, Java, C++

Technologies – Raft, Paxos, gRPC, Apache Kafka, Cassandra, PostgreSQL, etcd, ZooKeeper, Elasticsearch, Prometheus, Jaeger, Kubernetes, Docker

Tools – Git, Docker, Kubernetes, Terraform, GitLab CI/CD, DataDog, New Relic, Grafana, Protocol Buffers, Apache Flink, Redis